Inorganic Cume
April 25, 2014

The anion $Mo_2Cl_8^{-4}$ is known to have a Mo-Mo bond length of 2.14Å. The four chlorine atoms on each Mo are bonded approximately at right angles to the Mo-Mo bond axis and are in an eclipsed geometry. It may be prepared according to the following reactions:

$$Mo(CO)_6 \xrightarrow[\text{Acetic Acid}]{\text{diglyme}} Mo_2(O_2CCH_3)_4 \xrightarrow[0^oC]{HCl(aq.)} Mo_2Cl_8^{-4}$$

1) Describe the bonding in this molecule. Keep in mind that the molecule is diamagnetic.

2) Account for the difference in Mo-Mo bond lengths observed in the following complexes.

   $Mo_2(SO_4)_4^{-4}$, 2.11Å; $Mo_2(SO_4)_4^{-3}$, 2.16Å; $Mo_2(HPO_4)_4^{-2}$, 2.23Å

3) How would you have gone about determining whether or not a Mo-Mo bond existed in this molecule if you had been the first one to prepare it. (2 or more ways which need not be completely independent.)